Tree Traversal Practice

Data Structures

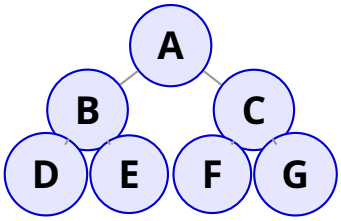
Problem 1: Tallest and Shortest Trees

Draw a binary tree with exactly 8 nodes that is as **tall** as possible. Then draw a binary tree with exactly 8 nodes that is as **short** as possible.

Problem 2: Full and Complete Trees

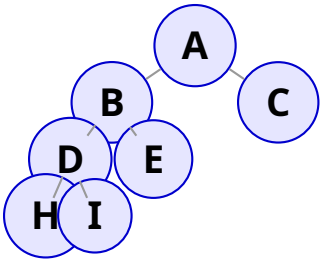
For each tree, mark whether it is **full** and whether it is **complete**.

Tree A



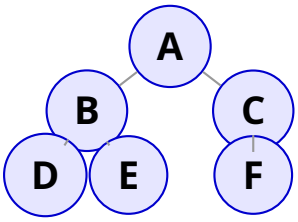
Full: Yes / No **Complete:** Yes / No

Tree B



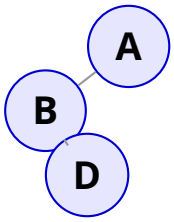
Full: Yes / No **Complete:** Yes / No

Tree C



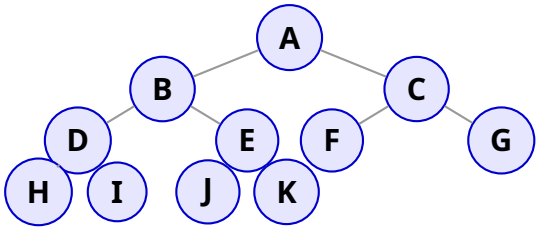
Full: Yes / No **Complete:** Yes / No

Tree D



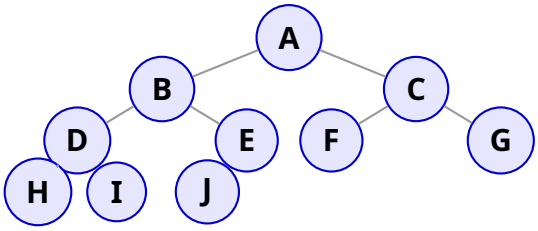
Full: Yes / No **Complete:** Yes / No

Tree E



Full: Yes / No **Complete:** Yes / No

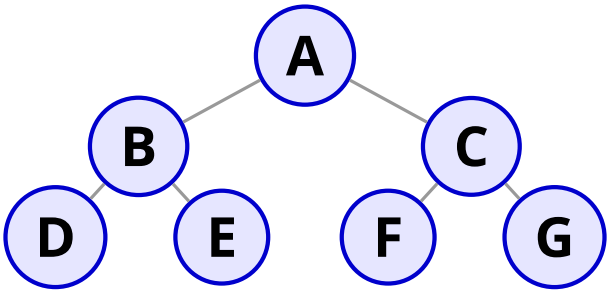
Tree F



Full: Yes / No **Complete:** Yes / No

Problem 3: Traversals of a Binary Tree

For each tree below, write the **inorder**, **preorder**, and **postorder** traversals.

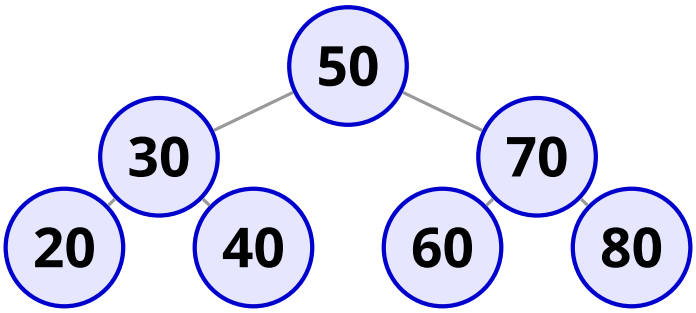


Inorder: _____

Preorder: _____

Postorder: _____

Tree 2



Inorder: _____

Preorder: _____

Postorder: _____

Answer Key

Problem 1: Tallest and Shortest Trees

The tallest tree is a chain of 8 nodes (7 edges from root to leaf). The shortest tree has 4 levels (height 3 edges); for example, fill the first three levels and place the eighth node as a child on the fourth level.

Problem 2: Full and Complete Trees

Tree	Full?	Complete?
A	Yes	Yes
B	Yes	No
C	No	Yes
D	No	No
E	Yes	No
F	No	Yes

Problem 3: Traversals

Tree	Inorder	Preorder	Postorde
1	D, B, E, A, F, C, G	A, B, D, E, C, F, G	D, E, B, F,
2	20, 30, 40, 50, 60, 70, 80	50, 30, 20, 40, 70, 60, 80	20, 40, 30